

HW1 for Modern Probability

Total Points: 100

For this set of problems, you may use results from real analysis without proof.

1. Let $\Omega = \mathbb{R}$ (or any uncountable set). Let

$$\mathcal{S} = \{\{\omega\} : \omega \in \Omega\}$$

be the family of singletons. Let

$$\mathcal{C} = \{A \subset \Omega : A \text{ or } A^c \text{ is countable}\}.$$

- (a) [5 points] Show that $\mathcal{C}$ is a σ -algebra.
(b) [5 points] Show that $\mathcal{C} = \sigma(\mathcal{S})$.

Remark: when $\Omega = \mathbb{R}$, $\mathcal{C} \neq \mathcal{B}(\mathbb{R})$.

[Total for Question 1: 10 points]

2. [10 points] Recall that $\mathcal{B}(\mathbb{R})$ is the σ -algebra generated by the sets $(-\infty, a]$ with $a \in \mathbb{R}$.

Let

$$\mathcal{Q} = \{(-\infty, q] : q \in \mathbb{Q}\}.$$

Show that

$$\sigma(\mathcal{Q}) = \mathcal{B}(\mathbb{R}).$$

3. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, and $A_n \in \mathcal{F}$, $n = 1, 2, \dots$. Let

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k, \quad \limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k.$$

- (a) [5 points] Show that

$$\mu\left(\liminf_{n \rightarrow \infty} A_n\right) \leq \liminf_{n \rightarrow \infty} \mu(A_n).$$

- (b) [5 points] Show that if $\mu(\Omega) < \infty$, then

$$\mu\left(\limsup_{n \rightarrow \infty} A_n\right) \geq \limsup_{n \rightarrow \infty} \mu(A_n).$$

[Total for Question 3: 10 points]

4. Recall two definitions:

We say that $\mathcal{A}$ is a π -system if it is closed under intersection, that is $A, B \in \mathcal{A} \implies A \cap B \in \mathcal{A}$.

We say that $\mathcal{D}$ is a λ -system if

- $\Omega \in \mathcal{D}$,
- $A, B \in \mathcal{D}, A \subset B \implies B \setminus A \in \mathcal{D}$,
- $A_n \uparrow A, A_n \in \mathcal{D} \implies A \in \mathcal{D}$.

Since any intersection of λ -systems is still a λ -system, we can define the *smallest* λ -system generated by an arbitrary collection of sets $\mathcal{A}$, denoted by $\lambda(\mathcal{A})$.

- (a) [5 points] Show that $\mathcal{A}$ is a σ -algebra if and only if it is both a π -system and a λ -system.
- (b) [5 points] Use the *method of appropriate sets* to show that if $\mathcal{A}$ is a π -system, then $\sigma(\mathcal{A}) = \lambda(\mathcal{A})$.

[Total for Question 4: 10 points]

5. (Completion of a probability space)

Let $(\Omega, \mathcal{F}_0, P_0)$ be a probability space. We say that $A \subset \Omega$ is a P_0 -null set (which may or may not be an element of $\mathcal{F}_0$), if there exists $N \in \mathcal{F}_0$ such that $A \subset N$ and $P_0(N) = 0$. Denote by $\mathcal{N}$ the collection of all P_0 -null sets.

- (a) [3 points] Let

$$\mathcal{F} = \{A \subset \Omega : \exists B_1, B_2 \in \mathcal{F}_0 \text{ s.t. } B_1 \subset A \subset B_2, A \setminus B_1, B_2 \setminus A \in \mathcal{N}\}.$$

Show that $\mathcal{F}$ is a σ -algebra, and show that it is the smallest σ -algebra containing $\mathcal{F}_0$ and $\mathcal{N}$.

- (b) [3 points] Let $P : \mathcal{F} \rightarrow [0, 1]$ be defined by $P(A) = P_0(B_1)$ where $B_1 \subset A, A \setminus B_1 \in \mathcal{N}$ and $B_1 \in \mathcal{F}_0$. Show that this definition is independent of the choice of B_1 .
- (c) [4 points] Show that $(\Omega, \mathcal{F}, P)$ is a probability space.
- Remark: this is called the completion of $(\Omega, \mathcal{F}_0, P_0)$.*

[Total for Question 5: 10 points]

6. Let μ be a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We say that $x \in \text{supp } \mu$ (the support of μ), if $\mu(x - \varepsilon, x + \varepsilon) > 0$ for every $\varepsilon > 0$.

- (a) [5 points] Show that if $\mu\{x\} > 0$, then $x \in \text{supp } \mu$.
- (b) [5 points] Show that if $\mu = \mu_X$ is the distribution of a continuous r.v. X with continuous density f , and $f(x) > 0$, then $x \in \text{supp } \mu$.
- (c) [5 points] Show that $(\text{supp } \mu)^c$ is an open set, that is, if $x \notin \text{supp } \mu$, then there is $\delta > 0$ such that $(x - \delta, x + \delta) \cap \text{supp } \mu = \emptyset$. As a consequence, $\text{supp } \mu$ is always a closed set.

- (d) [5 points] Recall the the Cantor set $\mathcal{C}$ is obtained by removing the middle intervals $I_n^{(k)}$, $1 \leq k \leq 2^{n-1}$, iteratively. Let μ_φ be the measure generated by the Cantor function. Show that $I_n^{(k)} \cap \text{supp } \mu_\varphi = \emptyset$ for every n, k , and deduce that $\text{supp } \mu_\varphi = \mathcal{C}$.

[Total for Question 6: 20 points]

7. Enumerate $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, q_3, \dots\}$ A Dirac measure δ_x on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is defined by

$$\delta_x(A) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Consider the probability measure

$$\mu = \sum_{n \geq 1}^{\infty} 2^{-n} \delta_{q_n}.$$

- (a) [3 points] Find all atoms of μ .
 (b) [3 points] Find the support of μ .
 (c) [4 points] Find the set of all discontinuity points of its cumulative distribution function.

[Total for Question 7: 10 points]

8. Let $f, g \geq 0$ be two Borel measurable and locally Lebesgue integrable functions. Let

$$\mu(A) = \int_A f(x) dx, \quad \nu(A) = \int_A g(x) dx.$$

- (a) [5 points] Show that $\mu \ll \nu$ iff $f = 0$ a.e. on $\{g = 0\}$.
 (b) [5 points] Show that $\mu \perp \nu$ iff $f(x)g(x) = 0$ for Lebesgue-a.e. x .

[Total for Question 8: 10 points]

9. [10 points] Prove or give a counterexample for the following statement concerning measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

- a) $\mu \perp \text{Leb}$ implies μ has atoms.
 b) $\mu \perp \text{Leb}$ implies μ has no atoms.
 c) μ has no atoms implies $\mu \ll \text{Leb}$.
 d) $\mu \ll \text{Leb}$ implies μ has no atoms.